

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Klastoets 1
VRAESTEL
Kopiereg voorbehou

Class Test 1
QUESTION PAPER
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Module EBN122
Elektrisiteit en Elektronika
August 2017

Module EBN122
Electricity and Electronics
August 2017



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:
Exam information:

MEMO

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form 30 min + 10 min om OKH vorm te voltooi	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	8 + OCR		

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session.

Do step 1 in the first 30 minutes of the test

Do step 2 within the following 5 minutes.

STEP 0 (0 min to 5 min): Complete the front page of the OCR form.

STEP 1 (5 min to 35 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (35 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2017EBN122C1**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	–	1	.	4	5					m	A								
02	3	4	0							V									
03	–	5	.	2															
04	3	1								k	Ω								

INSTRUKSIES

Doen stap 0 in die eerste 5 minute van die sessie.

Doen stap 1 in die eerste 30 minute van die toets.

Doen stap 2 binne die daaropvolgende 5 minute.

STAP 0 (0 min to 5 min): Voltooi die voorblad van di OKH vorm

STAP 1 (5 min to 35 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (35 min to 40 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.

- Die "Assessment ID" is **2017EBN122C1**
- Sluit die korrekte eenheid vir alle antwoorde in. Maak gebruik van die toepaslike voorvoegsel.
- Gebruik 'n meganiese potlood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak eerder gebruik van punte.

Voorbeelde

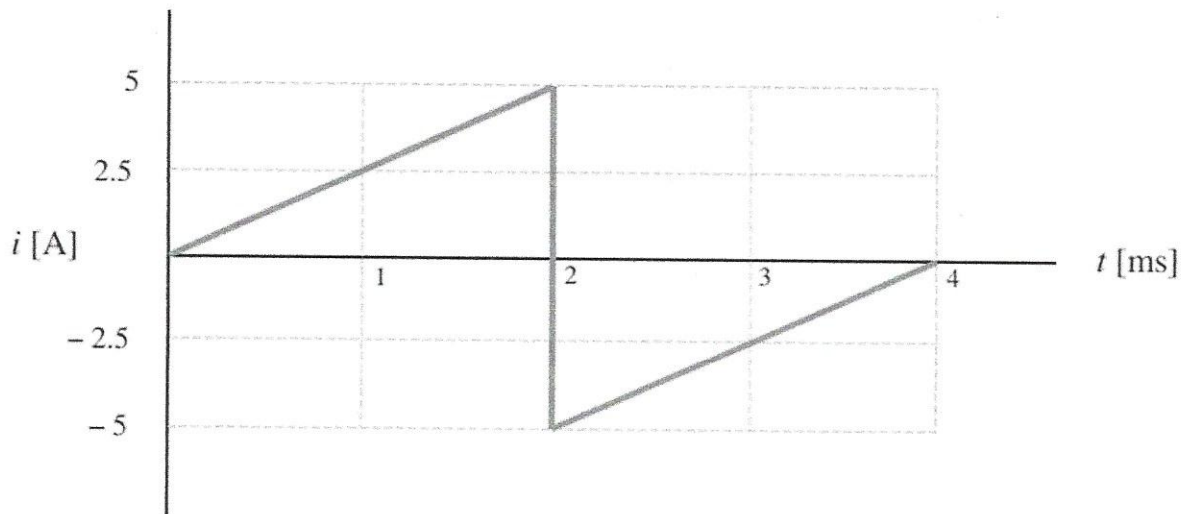
01	–	1	.	4	5					m	A								
02	3	4	0							V									
03	–	5	.	2															
04	3	1								k	Ω								

Question 1 & 2 [4]		Vraag 1 & 2 [4]	
01	A pie shop uses a 1300 W microwave to heat up the pies once they are sold. If it takes 2 minutes per pie to warm up, and 90 pies are sold in a day, what is the electricity cost for the microwave per day? Electricity costs R5/kWh.	01	A winkel wat pastye verkoop gebruik 'n 1300 W mikrogolfoond om die pastye warm te maak. Indien dit 2 minute per pasty neem om warm te word, en daar 90 pastye per dag verkoop word, wat is die elektrisiteitskoste van die mikrogolf per dag? Elektrisiteit kos R5/kWh.
02	A rechargeable flashlight battery of 5 V is capable of delivering a constant 100 mA. How long will the battery last if it is capable of delivering 15 kJ of energy when fully charged?	02	'n Herlaaiibare flitsligbattery van 5 V kan 'n konstante 100 mA lewer. Hoe lank sal die battery hou indien dit 15 kJ se energie kan lewer wanneer dit vol gelaaai is?

$$\begin{aligned}
 1) \text{ Cost} &= 1,3 \text{ kW} \times \text{R}5/\text{kWh} \times \frac{2}{60} \text{ h} \times 90 \\
 &= \text{R}19,50 \rightarrow
 \end{aligned}$$

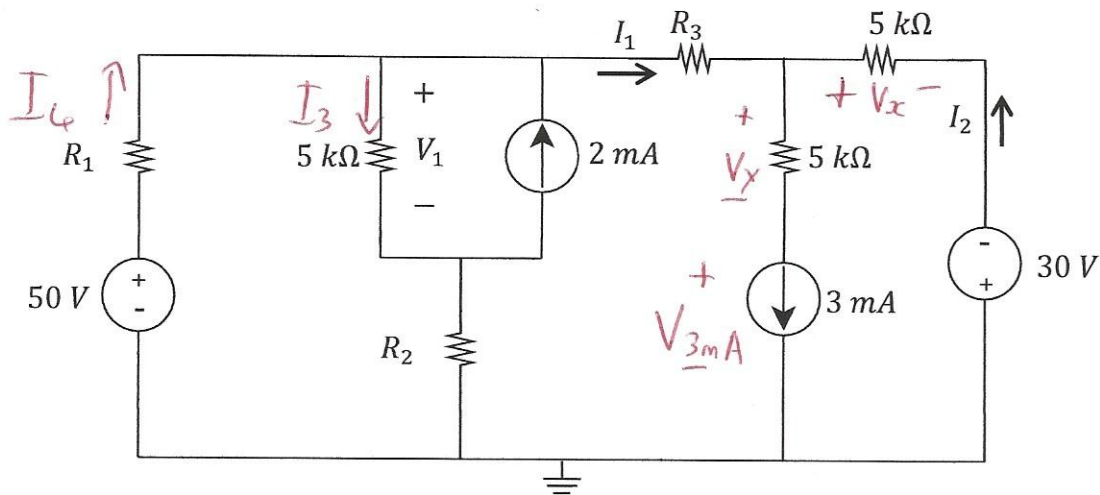
$$\begin{aligned}
 2) \text{ } w &= p t \\
 &= V I t \\
 t &= \frac{w}{V I} \\
 &= \frac{15 \text{ k}}{(5)(100 \times 10^{-3})} \\
 &= 30000 \text{ s} \rightarrow
 \end{aligned}$$

Question 3 [2]		Vraag 3 [2]	
03	The change in the current through an element is shown in the graph below. Determine the total charge through the element between 0 and 3 ms.	03	Onderstaande grafiek dui die verandering van die stroom deur 'n element aan. Bepaal die totale lading deur die element tussen 0 en 3 ms.



$$\begin{aligned}
 Q &= \frac{1}{2}(2 \times 10^{-3})(5) + (1 \times 10^{-3})(-2,5) + \frac{1}{2}(1 \times 10^{-3})(-2,5) \\
 &= 0,005 - 0,0025 - 0,00125 \\
 &= 1,25 \text{ mC}
 \end{aligned}$$

Question 4 & 5 [4]		Vraag 4 & 5 [4]	
If $I_1 = 1 \text{ mA}$, $I_2 = 2 \text{ mA}$ and $V_1 = 10 \text{ V}$, calculate the power associated with the 3 mA current source and the 50 V voltage source.		Indien $I_1 = 1 \text{ mA}$, $I_2 = 2 \text{ mA}$ en $V_1 = 10 \text{ V}$, bepaal die drywing geassocieer met die 3 mA stroombron en die 50 V spanningsbron.	
04	$P_{3\text{mA}}$	04	$P_{3\text{mA}}$
05	$P_{50\text{V}}$	05	$P_{50\text{V}}$



$$V_{ox} = -(2 \times 10^{-3})(5 \times 10^3) = -10 \text{ V}$$

$$V_y = (3 \times 10^{-3})(5 \times 10^3) = 15 \text{ V}$$

$$\text{KVL: } -V_{3\text{mA}} - V_y + V_x - 30 = 0$$

$$\therefore V_{3\text{mA}} = -15 - 10 - 30 = -55 \text{ V}$$

$$P_{3\text{mA}} = +(-55)(3 \times 10^{-3}) = -165 \text{ mW}$$

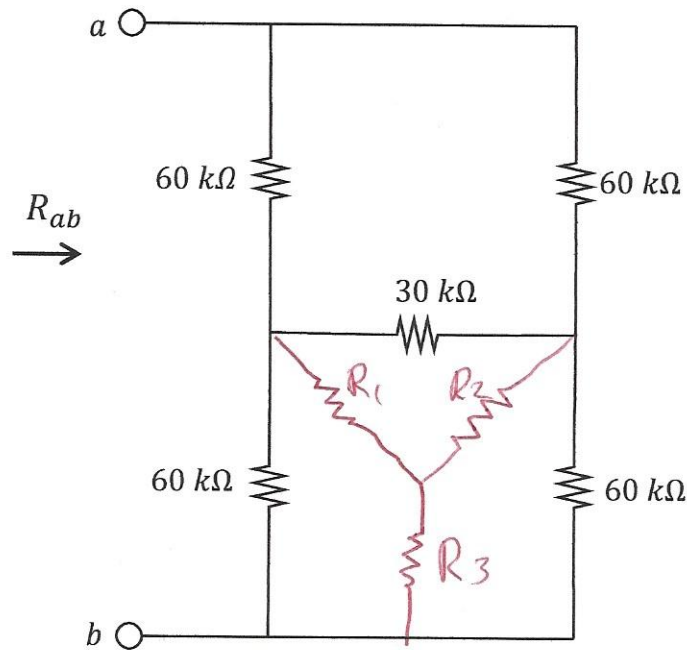
$$I_3 = \frac{10}{5\text{k}} = 2 \text{ mA}$$

$$\text{KCL: } I_4 + 2 \text{ mA} = I_3 + I_1$$

$$I_4 = 2 \text{ mA} + 1 \text{ mA} - 2 \text{ mA} = 1 \text{ mA}$$

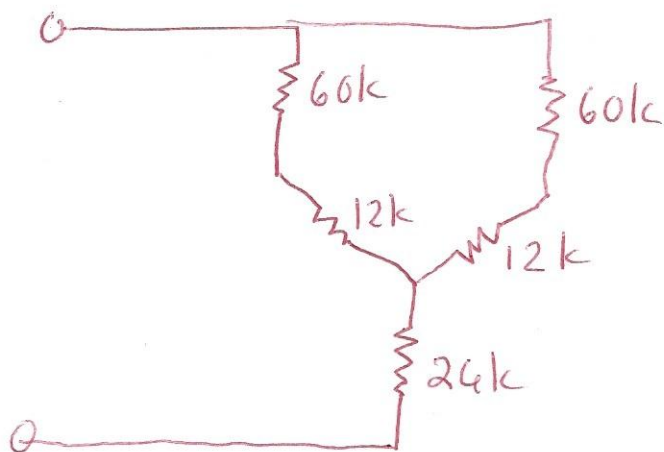
$$\therefore P_{50\text{V}} = -(50)(1 \times 10^{-3}) = -50 \text{ mW}$$

Question 6 [2]		Vraag 6 [2]	
06	Determine R_{ab} .	06	Bepaal R_{ab} .



$$R_1 = R_2 = \frac{(30k)(60k)}{30k + 60k + 60k} = 12k\Omega$$

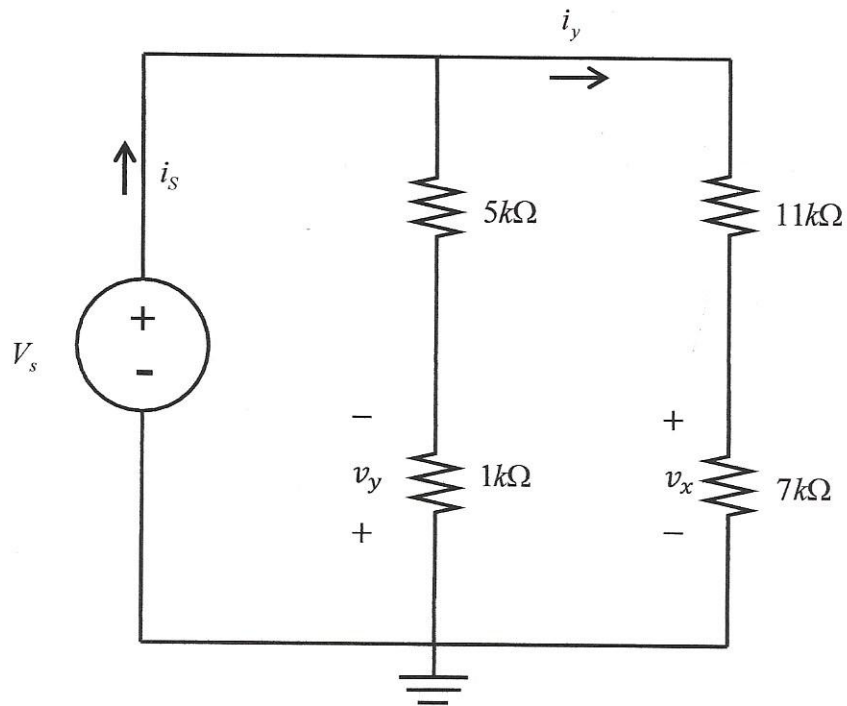
$$R_3 = \frac{(60k)(60k)}{30k + 60k + 60k} = 24k\Omega$$



$$\therefore R_{ab} = \frac{(60k + 12k)}{2} + 24k$$

$$= 60k\Omega$$

Question 7 & 8 [4]		Vraag 7 & 8 [4]	
07	Calculate V_S if $v_y = 5$ V.	07	Bepaal V_S indien $v_y = 5$ V.
08	Calculate v_x if $i_s = 10$ mA.	08	Bepaal v_x indien $i_s = 10$ mA.



$$V_y = \frac{1k}{1k + 5k} V_S$$

$$\therefore V_S = (-5) \left(\frac{6k}{1k} \right) = -30V$$

$$i_y = \frac{6k}{6k + 18k} (10mA)$$

$$= 2,5 mA$$

$$V_{oc} = (7 \times 10^3) (2,5 \times 10^{-3})$$

$$= 17,5 V$$

PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	7	E	B	N	1	2	2	C	1		
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Answer

Unit

01	$Cost =$	19,5	R
02	$t =$	30 000	s
03	$Q =$	1,25	mC
04	$P_{3mA} =$	-165	mW
05	$P_{50V} =$	-50	mW
06	$R_{ab} =$	60	k Ω
07	$V_S =$	-30	V
08	$v_x =$	17,5	V